# MILITARY SPECIFICATION

# INDICATOR, COMBUSTIBLE GAS AND TOXIC GAS

## 1. SCOPE

* 1.1 Scope. This specification covers two types of combustible gas and toxic gas indicators.

* 1.2 Classification. Indicators covered by this specification shall be of the following types as specified (see 6.2).

Type I - Indicator, combustible gas and toxic gas, without calibrator.

Type II - Indicator, combustible gas and toxic gas, with calibrator.

## 2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

| | |
|---|---|
| QQ-P-416 | Plating, Cadmium (Electrodeposited) |
| PPP-B-585 | Boxes, Wood, Wirebound |
| PPP-B-591 | Boxes, Fiberboard, Wood-Cleated |
| PPP-B-601 | Boxes, Wood, Cleated-Plywood |
| PPP-B-621 | Boxes, Wood, Nailed And Lock-Corner |
| PPP-B-636 | Box, Fiberboard |
| PPP-T-60 | Tape: Pressure-Sensitive Adhesive, Waterproof, For Packaging |

Military

| | |
|---|---|
| MIL-B-18 | Batteries, Dry |
| MIL-P-116 | Preservation, Methods of |
| MIL-D-1000 | Drawings, Engineering And Associated Lists |
| MIL-T-5624 | Turbine Fuel, Aviation, Grades JP-4 And JP-5 |
| MIL-P-7105 | Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements For |



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| | |
|---|---|
| MIL-S-7742 | Screw Threads, Standard, Optimum Selected Series: General Specification For |
| MIL-A-8625 | Anodic Coatings, For Aluminum And Aluminum Alloys |
| MIL-L-10547 | Liners, Case And Sheet, Overwrap; Water-Vaporproof Or Waterproof, Flexible |
| MIL-F-25524 | Fuel, Aircraft Turbine And Jet Engine, Thermally Stable |
| MIL-F-25558 | Fuel, Ramjet Engine, Grade RJ-1 |
| MIL-P-25576 | Propellant, Kerosene |
| MIL-J-25656 | Jet Fuel, Grade JP-6 |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-129 | Marking For Shipment And Storage |
| MIL-STD-130 | Identification Marking Of US Military Property |
| MIL-STD-143 | Specifications And Standards, Order Of Precedence For The Selection Of |
| MIL-STD-810 | Environmental Test Methods |

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction testing. This specification makes provision for preproduction testing.

3.2 Component parts. The indicator shall consist of a meter, a combustion chamber, and heated platinum wire(s) or other metallic filament(s) associated with a Wheatstone bridge type electrical circuit.

* 3.2.1 One calibrator per the number of Type II indicators specified in the contract shall be supplied with the indicator(s). The calibrator shall be considered a component of the Type II indicator. Calibrators shall be supplied as Aerospace Ground Equipment with Type I indicators as specified in the contract. The calibrator shall be a pocket size unit using heptane gas cartridges as the standard gas.

3.3 Materials.

3.3.1 Selection of specifications and standards. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided in the following paragraph:



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3.1.1 Standard parts. AN standard and MIL standard parts shall be utilized whenever suitable. They shall be identified by applicable part number. Commercial parts such as screws, bolts, nuts, and cotter pins may be used if replaceable by AN standard and MIL standard parts without alteration. The corresponding AN standard or MIL standard part number shall be on applicable drawings and parts lists. On applications for which no suitable AN standard or MIL standard is in effect on date of invitation for bids, commercial parts may be used provided requirements of this specification are adhered to.

3.3.2 Metals. Metals shall be of the corrosion-resistant type or be suitably treated to resist corrosion during normal service life. Materials contained within hermetically sealed inclosures are considered to be suitably protected from corrosion. Requirements specified for fungicidal and corrosion protective treatment, anodizing of aluminum alloy parts, and cadmium plating of steel parts are not applicable for parts within hermetically sealed instruments. Steel parts that have been cadmium plated and aluminum alloy parts that have been anodized shall be further protected against corrosion if necessary.

3.3.3 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the indicator, except where magnetic materials are essential.

3.3.4 Fungus-proof materials. Materials which are not nutrients for fungi shall be used to the greatest extent practicable. In cases where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent as approved by the procuring agency.

3.5 Protective treatment. When materials are used in the construction of the indicator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The usage of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Design and construction. The indicator shall be designed and constructed so that no parts will work loose in service and shall be built to withstand the strains, jars, vibration, and other conditions incident to shipping, storage, installation, and service without failure.

3.4.1 The construction of the indicator shall be reasonably simple to permit repair without requiring special tools or fixtures and without disassembly of the sampling or electrical circuits.

* 3.4.2 The indicator shall have two ranges of sensitivity. The upper range (least sensitive range) shall be designed to be sensitive in detecting small quantities of explosive vapors or gases associated with fuel oils or gasoline (including leaded gasoline), alcohol vapors, and hydrogen in mixtures with air, and shall indicate by means of the meter and by means of a conversion chart, whether the mixture is in the explosive range or below the lower explosive limit (LEL). The lower range (most sensitive range) shall be designed to be sensitive in detecting small quantities of hydrocarbon vapors or gases



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associated with fuel oils or gasoline and alcohol vapors in mixtures with air and shall indicate by means of the meter or a conversion chart, or both, whether the mixture is above the toxic level as established.

3.4.3 The indicator shall be designed to operate as specified herein, at any ambient temperature from -20° to 71° Centigrade (C).

3.5 Performance. The indicator shall be capable of meeting the tests specified in Section 4.

3.6 Details of components.

3.6.1 Combustion chamber and sensitive elements. The indicator shall be designed to permit operation in inclined position up to 90 degrees from normal operating position (instrument set to read zero in the inclined position with samples of atmospheric air that contains no combustible vapors), without affecting the accuracy of the indicator.

3.6.2 The combustion chamber shall be provided with replaceable flame arresters in the inlet and outlet ports to prevent flashbacks to the outside gases.

3.6.3 Aspirator. An aspirator bulb or small hand pump with restricted flow or check valve shall be provided for drawing a sample of the atmosphere under test into and through the combustion chamber. The restricted flow or check valve shall dampen the pumping impulses to prevent meter needle oscillations.

3.6.4 Sampling hoses. Two flexible sampling hoses of 5 and 15 foot lengths, each with suitable attachments for joining additional lengths of hose, shall be provided. The hoses shall be such that they will not cause indicating errors when utilized. Teflon lined neoprene hoses may be utilized.

* 3.6.5 Sample connections. Air tight fittings shall be attached to the indicator to allow for the attachment of the sampling hoses to the indicator. The external fitting connections shall be located so as not to interfere with the closing of the top cover when the sampling hoses are attached.

3.6.6 Controls.

3.6.6.1 The indicator shall be provided with a control or controls for obtaining the initial zero setting of the indicator and shall have a control or controls for changing the sensitivity (ranges) of the indicator.

* 3.6.6.2 Electrical controls. The control or controls for obtaining the initial zero setting on the indicator shall be located near the indicating dial to facilitate operation, and shall be constrained or free-wheeling so that the setting cannot be accidentally disturbed.

* 3.6.7 Batteries. The detector shall operate on dry batteries of type BA 30/U designated in MIL-B-18. The detector shall also operate on mercury or alkaline type batteries having the same configuration as type BA 30/U dry batteries.



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3.8 Battery voltmeter. The indicator shall include a voltmeter across the wheatstone bridge circuit or operating instructions for determining, without additional equipment, when the useful life of the batteries has been expended.

3.6.9 Indicator case. The indicator case shall be furnished with a hinged top cover, made of the same material as the case, to provide protection to the indicator while being stored or carried. This case may be constructed of metal, plastic, or any other suitable material and shall be properly treated or finished to resist corrosion. The top cover provided shall be arranged so that it cannot be completely closed unless the battery circuit is open. The material of the indicator case shall be nonmagnetic. The case shall be water-tight against rain.

* 3.6.10 Carrying strap. The indicator shall be equipped with a shoulder type carrying strap or other arrangement which will permit the indicator to be hung about the neck, leaving both hands free to operate the controls. The carrying strap shall be adjustable and shall be at least one half inch wide.

* 3.6.11 Calibrators. The calibrator shall check the accuracy of both ranges. The calibrator shall be such that the combustible gases supplied by the calibrator to the indicator will not be more than their indicated amount in terms of parts per million (PPM) or LEL of the calibration gas-air mixture, no less than 50 PPM of the indicated amount on the PPM range and no less than 5 percent LEL of the indicated LEL amount on the LEL scale. The calibration gas-air mixture need not be a Heptane-Air Mixture; however, conversion charts shall be provided convert to terms of a Heptane-Air Mixture. The design of the calibrator all be such that its value will not change because of usage or age. If the value of the calibrator varies as a result of environmental changes (temperature, humidity, altitude, etc.), correction charts shall be supplied with the calibrator. All calibrators shall be calibrated before they are shipped to the Government, the value of each and its expected accuracy shall be permanently affixed to the calibrator. If ampoules containing air-gas mixtures are utilized in the calibrators, their contents shall be indicated on the ampoules, as well as the accuracy of the air-gas mixture. The calibrators shall also calibrate the following instruments:

a. USAF Type R-1 Explosimeter P/N DN-74983
(Mine Safety Appliances Co.) FSN 6665-530-0985

b. Model Two Portable Explosimeter P/N DN-16386
(Mine Safety Appliances Co.) FSN 6665-490-2479

c. USAF Type R-2 Explosimeter P/N TBA 5S 10E1GA
(Industrial Instruments, Inc.) FSN 6665-978-3045

* 3.6.12 Filaments. Filaments shall be either of a plug-in or screw-in type and shall be capable of operating in 160 percent LEL conditions without damage.

* 3.6.13 Alarm. In addition to the meter, an audible alarm shall be included in the design; however, power for the alarm circuit shall be provided by a set of batteries separate from the indicator power source.



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* 3.7 Interchangeability. Parts having identical manufacturer's part numbers shall be completely interchangeable with respect to installation and performance. Manufacturer's part number changes shall be governed by the drawing number requirements of MIL-D-1000.

* 3.8 Scale ranges. The scale of the upper range of the meter shall cover the deflection produced for an air mixture of the gases and vapors, and shall consist of not less than 32 scale divisions reading from 0 to 160 percent of the LEL with figures at the 0, 20, 40, 60, 80, 100, 120, 140 and 160 percent graduations as shown on Figure 1. The scale of the lower range of the meter shall cover the deflection produced for an air mixture of the gases and vapors, and shall consist of not less than 32 scale divisions reading from 0 to 1600 PPM by volume, with figures at 0, 200, 400, 600, 800, 1000, 1200, 1400 and 1600 PPM by volume as shown on Figure 1. A heavy line (Figure 1) shall extend from the 500 PPM to the 1600 PPM mark. All scale graduations, all numerals, the dial pointer, and the heavy line (Figure 1) shall be painted red.

* 3.9 Accuracy. On the upper scale, the difference between the indicated concentration and the true concentration, expressed in percent of the LEL, of any gas or vapor shall not be less than the true concentration or exceed 30 percent of the true concentration. On the lower scale, the difference between the indicated concentration and the true concentration, expressed in percent of 1000 PPM, shall not be less than the true concentration nor exceed 30 percent of the true concentration.

* 3.10 Calibration. The indicator calibration reference shall be heptane. A standard conversion chart shall be provided to correct the indicated percent of the LEL to the true percent of the lower explosive limit and to correct the indicated PPM for each of the following gases and vapors:

a. Jet, rocket and reciprocating engine fuels:

(1) JP-4 (MIL-T-5624)

(2) JP-5 (MIL-T-5624)

(3) JP-6 (MIL-J-25656)

(4) JPTS (MIL-F-25524)

(5) RJ-1 (MIL-F-25558)

(6) RP-1 (MIL-P-25576)

(7) RP-2, Diethyleyelohexane

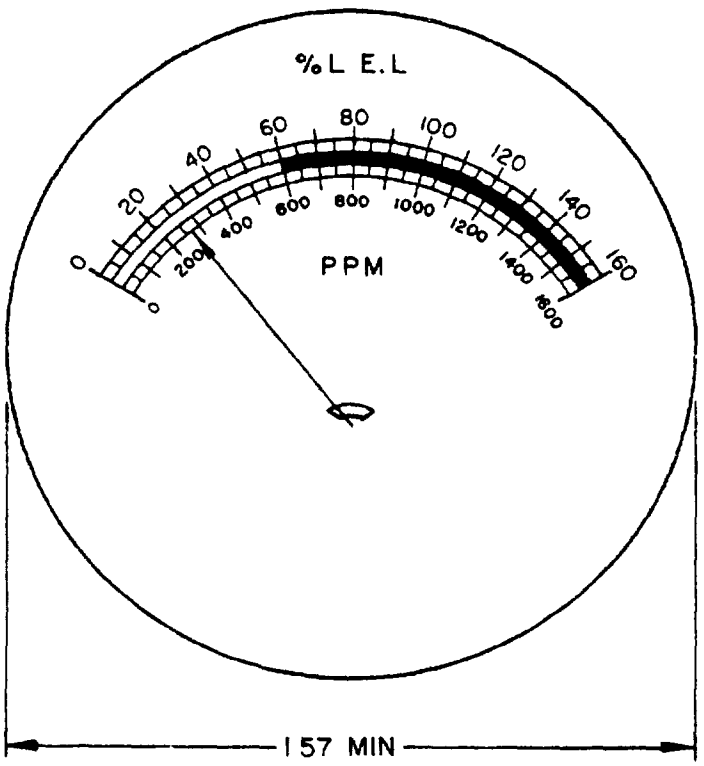
(8) Aviation gasoline of the following octanes:

(a) 80 to 87

(b) 91 to 96



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NOTE. METER MAY BE OF A SQUARE OR OBLONG TYPE IF BOTH ITS LENGTH AND ITS WIDTH ARE GREATER THAN 1 57 INCHES

# FIGURE 1 METER DIAL



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(c) 100 to 130

(d) 108 to 135

(e) 115 to 145

b. Ethyl alcohol

c. Hydrogen

d. Benzine

e. Methane

f. Hexane

g. Pentane

h. Heptane

3.11 Dimensions. The dimensions of the indicator shall be as shown on Figure 2.

3.12 Weight. The total weight of the indicator including batteries, but excluding the sampling hose, shall not exceed 9 1/2 pounds.

3.13 Threads.

3.13.1 Unless otherwise specified, the threads of all machine screws 0.060 inch or larger in diameter shall conform to MIL-S-7742.

3.13.2 Pipe threads shall conform to MIL-P-7105.

3.14 Finishes and protective coatings.

3.14.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except that the dial, small holes, and case inserts need not be anodized.

3.14.2 Steel parts. Steel parts shall be cadmium plated, where practicable, in accordance with QQ-P-416.

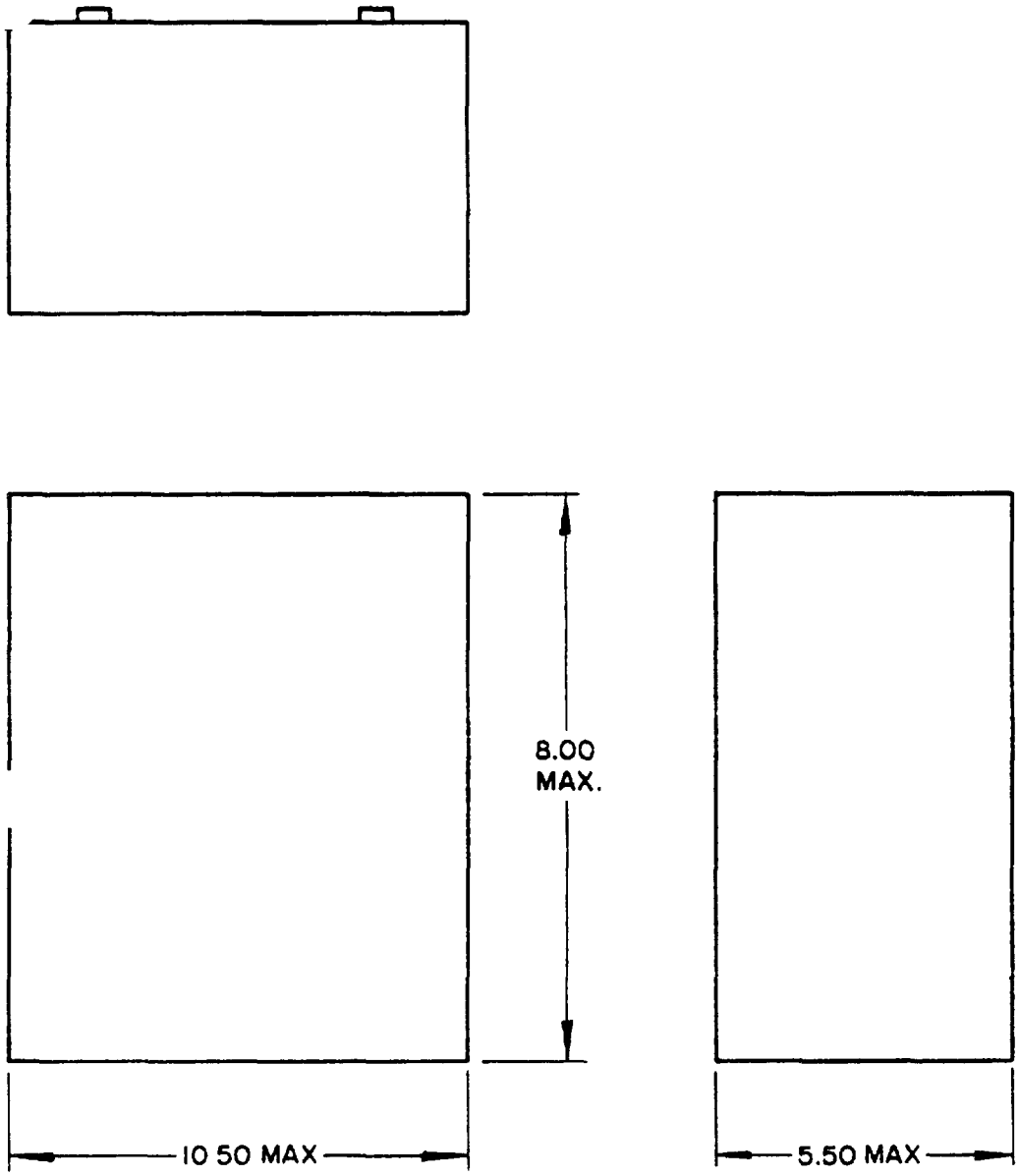
3.15 Marking.

3.15.1 All markings shall be durable to withstand the usage encountered in service. Numerals shall distinctly indicate the graduation to which each applies.

3.15.2 Visibility of dial. The pointer, numerals, at least 1/16 inch of the shortest graduations, and other specified markings on the dial shall be visible from any point within the frustum of a cone whose side makes an angle of



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DIMENSIONS IN INCHES

FIGURE 2 INDICATOR



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30 degrees with a perpendicular to the dial and whose small diameter is the aperture of the dial.

3.15.3 Operating instructions. A brief operating instruction shall be permanently and legibly mounted on the indicator case for ready reference.

3.15.4 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.16 Workmanship.

3.16.1 General. The indicator, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, painting, and wiring; alignment of parts and tightness of assembly screws and bolts; et cetera.

3.16.2 Cleaning. The indicator shall be thoroughly cleaned, and loose, spattered, or excess solder, metal chips, and other foreign material removed during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of indicators shall be classified as follows:

4.2.1 Preproduction tests. Preproduction tests are those tests performed on samples submitted for preproduction approval.

4.2.2 Inspection tests. Inspection tests are those tests accomplished on indicators submitted for acceptance under contract.

4.3 Test conditions.

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.



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.3.2 Tapping. Unless otherwise specified, the indicator shall be gently tapped or vibrated before a test reading is taken.

4.3.3 Attitude. Unless otherwise specified, the indicator shall be tested in normal operating position.

4.3.4 Vibration. A vibration stand shall be used which will vibrate at any desired frequency between 500 and 5000 cycles per minute (cpm.) and shall subject the indicator to vibration such that a point on the case will describe in a plane inclined 45 degrees to the horizontal plane, a circle, the diameter of which is equal to the amplitude specified herein.

* 4.4 Preproduction tests and samples.

4.4.1 Preproduction test samples. Preproduction test samples shall consist of 2 indicators representative of the production item. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

4.4.2 The preproduction test shall consist of the inspection tests, sampling tests, and the following tests:

4.4.2.1 At the conclusion of each of the following tests, the indicator shall be subjected to and shall meet the requirements specified in 4.5.2.3.

* '.4.2.2 Low temperature. The indicator, including the calibrator, shall be ubjected to a temperature of -54°C for a period of 24 hours. The temperature shall then be raised to -20°C for a period of 24 hours. While still at this temperature, the indicator shall be tested for compliance with 4.5.2.3. The indicator shall not be tested with heptane-air mixtures of 50 and 100 percent of the LEL at -20°C. The dry batteries of type BA-30/U (3.6.7) may be stored at standard atmospheric conditions and inserted in the indicator just before it is tested for compliance with 4.5.2.3.

* 4.4.2.3 High temperature exposure. The indicator, including the calibrator, shall be exposed to high temperature in accordance with Method 501, Procedure I of MIL-STD-810. While still at this temperature, the indicator shall meet the requirements specified for scale error in 4.5.2.3. There shall be no damage which would adversely affect subsequent operation.

4.4.2.4 Endurance. The indicator, utilizing a 5 foot sampling hose and equipped with type BA 30 batteries, shall be operated for 24 hours at 50 percent of the LEL with heptane and air. The indicator shall be serviceable for a continuous period of at least 4 hours without requiring replacement of batteries. The indicator shall be serviceable and accurate within the limits specified for scale error on both ranges at the end of the 24 hour period with and without 5 foot sampling hose. The indicator shall not require the replacement parts other than batteries during this 24 hour period.



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4.4.2.5 After the 24 hour test, the instrument utilizing the 15 foot sampling hose shall be operated for 4 hours at 500 PPM of heptane and air. At the end of the 4 hour period, the indicator shall be serviceable and accurate within the limits specified for scale error on both ranges with and without the 15 foot sampling hose.

4.4.2.6 Rain. The indicator shall be mounted for one hour with the dial in a vertical plane and one hour with the dial in a horizontal plane (dial face upward) in a rain test chamber. The external connections shall be made to the indicator in such a manner as to simulate installed conditions. The test shall be performed in accordance with Method 506, Procedure I of MIL-STD-810. After the test, the indicator shall be examined to ascertain that no water has entered the indicator.

* 4.4.2.7 Vibration failure. The indicator including the calibrator shall be subjected to vibration with an amplitude between .018 and .020 inch for a 3 hour period. The frequency shall be varied uniformly from 500 to 3000 cpm and return once each minute during this test. There shall be no looseness in the mechanism nor damage to any part of the indicator, including the calibrator, as a result of the vibration. The indicator, including the calibrator, shall be securely mounted on a vibration table during this test.

* 4.4.2.8 Humidity. The indicator, including the calibrator, shall be mounted in its normal operating position in a chamber. The test shall be performed in accordance with Method 507, Procedure I of MIL-STD-810.

* 4.4.2.9 Fungus. The indicator, including applicable external connections and the calibrator, shall be subjected to the action of fungi in accordance with Method 508, Procedure I of MIL-STD-810. After the test, the indicator including the calibrator, shall be examined to ascertain that no fungus growth has occurred. At the option of the inspector, tests in addition to the above may be made to determine that there has been no deterioration which would adversely affect subsequent operation.

* 4.4.2.10 Salt. The indicator, including the calibrator, shall be subjected to salt spray in accordance with Method 509, Procedure I of MIL-STD-810. An evidence of corrosion or discoloration of any part of the indicator or calibrator shall constitute failure of the indicator and the calibrator to pass this test.

* 4.4.2.11 Response. The indicator shall respond to a 50 percent heptane-air mixture within 5 seconds at standard temperature, utilizing a 15 foot sampling hose.

* 4.4.2.12 Explosive resistance. The indicator shall be so constructed as to pass the explosive atmosphere test as shown in Method 511.1, Procedure I of MIL-STD-810.

* 4.4.2.13 Compatibility. The calibrator shall be tested with the indicators listed in 3.6.11 under standard atmospheric conditions (temperature approximately 70 degrees Fahrenheit (F) and humidity from 5 to 95 percent). The



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contractor shall submit a detailed description of the methods and procedures used in the tests. The temperature, atmospheric pressure, and humidity existing during this testing shall be recorded. The calibrator shall calibrate all indicators subjected to this test as specified in this specification.

4.5 Inspection tests.

4.5.1 Unless otherwise specified, contractor's records of all inspection work and tests, giving the results of tests required to determine compliance with the requirements and tests specified herein, shall be kept complete and shall be available to the Government representative at all times. The record or report of inspection and tests shall be signed or approved by a responsible person specifically assigned by the contractor.

4.5.2 Individual tests. Each indicator shall be subjected to the following tests:

* 4.5.2.1 Examination of product. Each indicator, including the calibrator(s), shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

* 4.5.2.2 Stability. On each range, the indicator shall be operated on fresh air for 15 minutes, and "zeroed" by means of the proper controls. On each range, the indicator shall then be operated on a heptane-air mixture of known concentration and identity for 3 consecutive periods of 5 minutes each, while being quickly and accurately "rezeroed" on fresh air before each period of 5 minutes. The sampling hose may not be disconnected from the indicator while "zeroing." The variation in reading shall not be lower than the true concentration nor exceed the true concentration by more than 30 percent.

* 4.5.2.3 Scale error. On the upper range, the indicator shall be tested with heptane-air mixtures of 25, 50 and 100 percent of LEL and the indicated concentration shall not be less than the true concentration or more than 30 percent LEL greater than the true concentration. On the lower scale, the indicator shall be tested with heptane-air mixture of 250, 500, and 1000 PPM by volume, and the indicated concentration shall not be less than the true concentration. The indicator shall be operated in an inclined position of 45° at each of the aforementioned heptane-air mixtures and shall read the mixtures within the accuracy specified for the heptane-air mixtures mentioned above. Finally, at the discretion of the Government inspector, he may require the contractor to demonstrate to the satisfactory operation of any indicator (utilizing the aspirator or hand pump) to him.

* 4.5.2.4 Safety. The indicator shall be operated on air-hydrogen and air-acetylene mixture of maximum explosibility; namely, a mixture of 2 parts hydrogen to 5 parts air and a mixture of one part acetylene to 12 parts air. Flashback from the combustion chamber of the indicator shall not occur.

4.5.3 Sampling tests. Five indicators, which have been manufactured under essentially the same conditions, shall be selected at random from each lot of 500 or fraction thereof and subjected to the following tests:



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4.5.3.1 When sampling tests are specified on a number of indicators that are selected as representative of the lot, and one or more fail to meet the specified tests, acceptance of all indicators in the lot will be withheld until the extent and cause of failure are determined. For operational reasons, individual tests may be continued pending investigation of a sampling test failure, but final acceptance of the entire lot is contingent upon the inspector's decision regarding the over-all conformance of the lot with specification requirements. When corrective action has been accomplished, all necessary tests shall be repeated. If investigation indicates that the defects may exist on lots previously accepted, full particulars concerning the defects found, including recommendations for correction, shall be furnished to the contracting officer.

4.5.3.2 Vibration error. The indicator shall be securely mounted on the vibration stand so that it is in the normal operating position. The indicator shall be operated on its own battery and shall be subjected to vibration with an amplitude between .003 and .005 inch at frequencies of from 500 to 3000 cpm.

4.5.3.2.1 Pointer vibration. The maximum amplitude of the pointer vibration and the pointer variation shall not exceed 1/16 inch.

4.5.3.2.2 Accuracy. The reading shall not differ at any time by more than 5 percent from the reading obtained when the indicator is at rest (percent difference is as expressed in 3.9).

4.6 All parts, specimens, or assemblies destroyed in making tests required by this specification or drawings, to determine compliance with this specification or drawings, shall be in addition to the quantity specified in the contract or purchase order and shall be furnished without increasing the cost of the contract or order.

4.7 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

5. PREPARATION FOR DELIVERY

5.1 Levels of preservation and packaging.

5.1.1 Level A. Unless otherwise specified by the procuring activity, each indicator shall be preserved and packaged in accordance with MIL-P-116, Method II.

5.1.2 Level C. When level C packaging is specified, indicators shall be preserved and packaged in accordance with manufacturer's commercial practice.

5.2 Levels of packing.

5.2.1 Level A. Unless otherwise specified by the procuring activity, indicators packaged as specified in 5.1.1 shall be packed in export-type shipping



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containers conforming to PPP-B-591, PPP-B-621, PPP-B-601, PPP-B-636 (Type I or II, Class 2). As far as practicable, exterior shipping containers shall be of uniform size and shape, but of minimum cube and tare, consistant with protection required and contain identical quantities. The gross weight of wood shipping containers shall not exceed 200 pounds. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto. Containers shall be provided with a case liner conforming to MIL-L-10547. The case liner may be omitted when the container conforms to Class 2 of PPP-B-636 provided the center edge seams and manufacturer's joints are sealed with tape conforming to Type II, Class 1 of PPP-T-60.

5.2.2 Level B. Unless otherwise specified by the procuring activity, indicators packaged as specified in 5.1.1 shall be packed in domestic-type shipping containers conforming to PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621, or PPP-B-636 (Type I or II, Class 1). Box closure shall be in accordance with the applicable container specification or appendix thereto. The gross weight of wooden shipping containers shall not exceed 200 pounds. When fiberboard boxes are used, the weight and dimensions shall not exceed the limitations of Table I of PPP-B-636.

5.2.3 Level C. Unless otherwise specified by the procuring activity, indicators packaged as specified in 5.1.2 shall be adequately packed in a manner to insure carrier's acceptance and safe delivery to the receiving activity. Exterior containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

INDICATOR, COMBUSTIBLE GAS AND TOXIC GAS
Specification MIL-I-38715(USAF)
Manufacturer's Part Number
Name of Manufacturer

6. NOTES

6.1 Use. This portable combustible gas indicator covered by this specification is intended for testing air in fuel tanks and other enclosed spaces for the presence of dangerous concentrations of combustible gases or vapors.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Location and corditions for preproduction testing.

c. Applicable levels of preservation, packaging, and packing.

d. Quantity of unit packages to be included in each intermediate package.



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e. Quantity of calibrators to be procured.

f. Type (Type I or Type II, see 1.2).

* 6.3 4.5.2.3 and 3.9 do not contradict each other. For example, on the upper range, the indicator is acceptable if it reads a 25 percent LEL mixture (true concentration) within the range from 25 percent LEL to 55 percent LEL. On the lower range, the indicator is acceptable if it reads, for example, a mixture of 250 PPM (true concentration) within the range from 250 PPM to 550 PPM. All percentage figures in this specification pertaining to accuracy are to be calculated on the upper range as a percent of 100 percent LEL and on the lower range as a percent of 1,000 PPM.

* 6.4 In the first sentence of 3.9 the words "any gas or vapor" refer only to vapors and gases listed in 3.10 mixed with air. Also 3.9 does not apply to mixtures of combustible gases or vapors and air in which the concentration of combustible gases is too great to be combustible or explosive.

* 6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Review Activity:
Air Force - 82

Preparing Activity:
Air Force - 82

Project No. 6665-F220



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# SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No 119-R004

**INSTRUCTIONS**

This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

| ORGANIZATION | CITY AND STATE |
|---|---|
| | |

| CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT |
|---|---|---|
| | | $ |

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF YES IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

| SUBMITTED BY *(Printed or typed name and activity)* | DATE |
|---|---|
| | |

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4063 WHICH IS OBSOLETE

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